

# The Anatomy of a Website

## Browsers, your first HTML, and the boilerplate — In-Class Discovery Worksheet

Name: \_\_\_\_\_ Section: \_\_\_\_\_ Date: \_\_\_\_\_

**How to use this sheet:** Keep your **browser** and your **Terminal** open beside you. For every task, **attempt and write your answer *before* we discuss it**. Today we open up a real website, peek at the HTML behind it, and write our very first page.

### Part 1 — The Browser

**Q1.** Which **web browser** do you use most often? Why do you prefer it over the others?

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**Q2.** List at least **five web browsers** you know. Which **company** develops each one?

| Browser | Developed by |
|---------|--------------|
| _____   | _____        |
| _____   | _____        |
| _____   | _____        |
| _____   | _____        |
| _____   | _____        |

**Q3.** What **is** a browser? What sort of things do you generally **do** using a browser?

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**Q4.** Let's talk about **Chrome**. Have you heard of it? I'll be using it throughout the course. List **5 pros** of using Chrome.

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#### TRY THIS · CHROME

Make **Chrome** your **default browser**. (Settings → "Default browser" → *Make default*.) From now on, links you click will open in Chrome.

**Q5.** You already know about **domains** from earlier sessions. Open **three popular websites** using their domains in **three different tabs**. What do you see? (e.g. `youtube.com`, `flipkart.com`, `zenithschool.ai`)

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### TRY THIS - BROWSER

**Right-click** anywhere on the page — this opens a *context menu* — and click "**View Page Source**". Repeat this for **all three** websites.

**Q6.** What do you see here? Look at the **first few lines** of each — what looks **common** across all three?

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## Part 2 — First Interaction with HTML

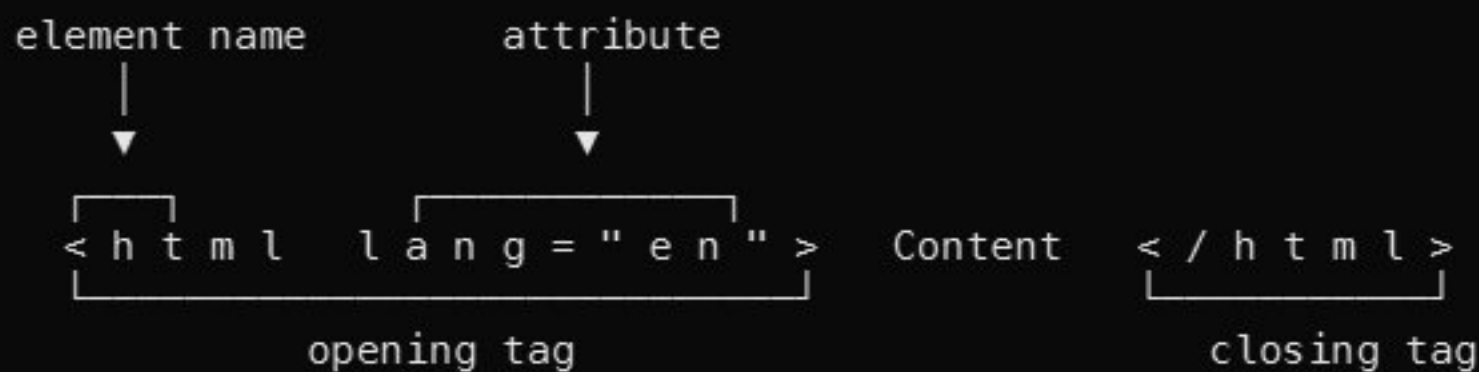
The output you just saw is **HTML**. The very first line, `<!DOCTYPE html>`, is a **document type declaration**. It tells the browser you are using **HTML5**, the latest version of HTML.

**Q7.** If I tell you that an **HTML tag** is enclosed within `<` and `>`, which HTML tags are you able to **identify** in the page source right now?

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We have different **HTML elements**. Each element usually has **4 main parts**: an **opening tag**, **attribute(s)**, **content**, and a **closing tag**.



**Opening Tag:** marks the *start* of the element — the element name in angle brackets (e.g. `<p>`).

**Attributes:** optional properties placed inside the opening tag to give extra info or configuration.

**Content:** the actual text, image, or media shown on the screen.

**Closing Tag:** marks the *end* of the element — identical to the opening tag but with a forward slash (e.g. `</p>`).

**Q8.** Now — what do you think is the difference between an **Element** and a **Tag**?

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**Q9.** Do you see any elements that have **no closing tag**? If yes, name them.

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### TRY THIS - TERMINAL

Let's create our own HTML page.

```
mkdir Full_Stack_Sessions # create a new folder
cd Full_Stack_Sessions
touch 1.html # create an empty file
vim 1.html # open it in vim
```

Inside `1.html`, type:

```
<h1>Hello!</h1><p>I am Likhilesh!</p>
```

Save and exit vim (`:wq`), then run:

```
open 1.html
```

**Q10.** What do you see? Open the **Page Source** and **relate it** to the output on screen.

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### TRY THIS - BROWSER

Open the context menu again (**right-click** anywhere on the page), but this time click **"Inspect"**. This opens the **Developer Tools** area, with tabs like **Elements**, **Console**, **Network**, **Application**, ... This area is built specially for developers — hence the name.

**Q11.** What all did you see in the Developer Tools? Is there a specific area where you see something that looks like **HTML**?

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**Q12.** Pay close attention: what is **different** between what you see in **"View Page Source"** and what you see in the **"Elements"** tab inside Developer Tools?

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## Part 3 — Understanding the HTML Boilerplate

The browser receives the initial **HTML** and reads it. Side by side, it builds a **tree-like structure** called the **DOM**. The *live view* of this DOM is exactly what the **Elements** tab shows. We'll talk a lot more about the DOM when we get to JavaScript.

Raw HTML —the browser reads it→ DOM (tree)

|                                                                                              |                                          |
|----------------------------------------------------------------------------------------------|------------------------------------------|
| <pre>&lt;html&gt;   &lt;head&gt;     &lt;title&gt;...   &lt;body&gt;     &lt;h1&gt;...</pre> | <pre>html  /  \ head body    title</pre> |
|----------------------------------------------------------------------------------------------|------------------------------------------|

**Q13.** How will you write your code when you build **big websites / web-applications**? How will you **manage** it? You've already heard of **Git** — what else?

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**Q14.** What **IDEs** have you heard of? What **problems** do they solve?

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**Q15.** We'll use **Visual Studio Code**, the gold standard of web development. Have you heard of it? Write **2 reasons** why I might call it the "Gold Standard".

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#### TRY THIS - VS CODE

Open **Visual Studio Code**. Click "**Open Folder**" and open the `Full_Stack_Sessions` folder you created earlier.

**Q16.** Do you see the `1.html` file present in the folder now?

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#### TRY THIS - VS CODE

Click on `1.html` on the left. **Remove all of its content** — make it blank. Now type `!` (an exclamation mark); a suggestion appears. Press **Enter** after you see the suggestion.

**Q17.** What do you see now? This is called the **Boilerplate code** of HTML. Write your observations here.

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**Q18.** You can see `html` is the **parent** element, while `head` & `body` are its **children**. Draw an **upside-down tree** diagram of this — start from `html` and include **everything** you see in the boilerplate.



### TRY THIS - VS CODE + BROWSER

In between the **opening and closing tag of the body element**, write the **same code** we wrote earlier in `1.html`, then **save** the file:

```
<h1>Hello!</h1><p>I am Likhilesh!</p>
```

In the left sidebar, **right-click** on the filename `1.html` and select **"Reveal in Finder"** (or "Reveal in File Explorer"). Your file manager opens — **double-click** the file there. It opens as a webpage in your **default browser**. (*I recommend making Chrome your default browser if you haven't already.*)

**Q19.** What is the difference between the **"Page Source"** you can see now compared to earlier? Is the **output on the screen** (the web page) different?

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**Q20.** Open the **"Elements"** tab in Developer Tools — does it look the **same** as earlier, or is there any **change**?

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### CLASS WRAP-UP - TERMINAL + GIT

In VS Code's left sidebar, use the **"New Folder"** button to create a folder named `1_HTML`, then **cut and paste** `1.html` into it — this is just an arrangement so you can organise your files conveniently in future sessions. Now link the root folder `Full_Stack_Sessions` to a new **"Public"** git repository (keep the GitHub repository name `Full_Stack_Sessions`), then **add, commit** and **push** your code to GitHub:

```
git init
git add .
git commit -m "First HTML page"
git push
```

### TRY THIS - GITHUB PAGES

Let's put your page on the internet. Open your GitHub repository and:

1. Go to the **Settings** tab (last item in the same row as the **Code** tab).
2. Click **Pages** in the left sidebar.
3. Under "**Deploy from a branch**", select **GitHub Actions**.
4. In the first card, "**GitHub Pages Jekyll**", click **Configure**.
5. Click the green "**Commit changes...**" button, then click "**Commit changes**" once more to confirm.
6. Go to the **Actions** tab (also at the end of the **Code** row). You'll see a deployment "**Queued / In progress**" (yellow). It completes in roughly **50 seconds** and the marker turns **green**.
7. Go back to **Settings** → **Pages** — a **link** now appears that wasn't there before. Open it.

**Q21.** Do you see any output on the webpage, or is it an **error page**?

### TRY THIS - BROWSER

After your link, type `1_HTML/1.html` — does this show any output? The link acts as a **base** that redirects to your repository. Your HTML file lives inside the `1_HTML` folder and is named `1.html`, so the full path is *link + folder + file name*:

```
<link>/1_HTML/1.html
```

**Q22.** Share this link with a friend on **WhatsApp** and ask them to open it and see the output.

**Note:** You made those changes on GitHub, which created a `.github` folder on the **remote** — but that folder is not yet on your local machine. So you need to **pull** those changes down:

```
git pull
```